

Show all your work in the space provided for each problem. All problems will be graded for correctness, completeness, conciseness, and clarity. Do your best. Don't cut corners.

1. (8 pts) List all the elements and the binary operation which make each set a group.

a) $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$

operation addition mod 6

b) $U(8) = \{1, 3, 5, 7\}$

operation multiplication mod 8

2. (4 pts) Write each of the following statements using the "divides" notation $x|y$

a is a divisor of b

$$a|b$$

a is a multiple of b

$$b|a$$

If p is a factor of n and x is any integer then the product nx is divisible by p.

$$\text{If } p|n \text{ and } x \in \mathbb{Z} \text{ then } p|nx.$$

3. (6 pts) Let G be a group with operation *. Use existential and universal quantifiers as needed to define the following properties for G.

a) The associative property $\forall x, y, z \in G \quad (x * y) * z = x * (y * z)$

b) The inverse property $\forall x \in G, \exists y \in G \text{ such that } x * y = e = y * x$

c) The commutative property $\forall x, y \in G \quad x * y = y * x$

14 4. (8 pts) What is the order of each of the following groups?

25 a) $|Z_n| = n$

19 b) If p is prime $|U(p)| = p - 1$

26 c) $|D_n| = 2n$

22 d) $|S_n| = n!$

5. (8 pts) Use the Cayley table given below for the group G with operation $\circ$ to answer the following questions.

	I	A	B	C	D	E	F	G
I	I	A	B	C	D	E	F	G
A	A	B	C	I	G	F	D	E
B	B	C	I	A	E	D	G	F
C	C	I	A	B	F	G	E	D
D	D	F	E	G	I	B	A	C
E	E	G	D	F	B	I	C	A
F	F	E	G	D	C	A	I	B
G	G	D	F	E	A	C	B	I

23 a) (2 pt) What is the inverse of A?

C

b) (2 pts) List all elements of order 2 in G .

15 B, D, E, F, G

c) (2 pts) List all elements in the center of G .

10 I, B

d) (2 pts) Prove that G is **not** Abelian.

25 $A \circ G = E$ But $G \circ A = D$ so G is not Abelian.

6. (12 pts) Let $\alpha = (1\ 3\ 5\ 7\ 9)(2\ 4)(6\ 8)$ and $\beta = (1\ 5\ 2\ 4)(6\ 7\ 8\ 9)$ be permutations in S_9 .

25 a) Write β in **array form**. $\beta = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 5 & 4 & 3 & 1 & 2 & 7 & 8 & 9 & 6 \end{bmatrix}$

20 b) Write the product $\alpha\beta$ in **standard cycle form**. $\alpha\beta = (1\ 7\ 6\ 9\ 8)(3\ 5\ 4)$
 $= (1\ 7\ 6\ 9\ 8)(3\ 5\ 4)$

24 c) What is the order of each element: $|\alpha| = 10$ $|\beta| = 4$
 $\text{LCM}(5, 2)$ $\text{LCM}(4, 4)$

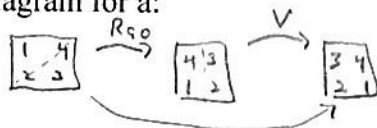
13 d) Write α^{154} in standard cycle form. $\alpha^{154} = \alpha^4 = (1\ 9\ 7\ 5\ 3)$

7. (6 pts) Shown below is a blank Cayley table for the group D_4 . Write the elements in cells labeled a and b as a product of elements in D_4 . Then draw diagrams to determine exactly which element should be in the cell. You must show your diagrams to receive full credit.

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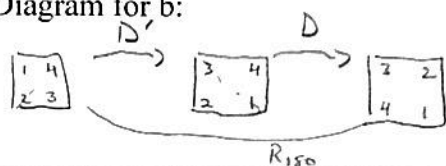
$$a = V \cdot R_{90} = D'$$

Diagram for a:



$$b = D \cdot D' = R_{180}$$

Diagram for b:



	R_0	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_0								
R_{90}								
R_{180}								
R_{270}								
H								
V		$V \circ R_{90}$ a						
D								$D \circ D'$ b
D'								



H



V



D



D'

8. (8 pts) Let $\sim$ be an equivalence relation on a set S . Prove that any two equivalence classes $[a]$ and $[b]$ are either disjoint or identical (either $[a] \cap [b] = \emptyset$ or $[a] = [b]$)

If $[a] \cap [b] = \emptyset$ then done. Show if $[a] \cap [b] \neq \emptyset$ then $[a] = [b]$.

Let $c \in [a] \cap [b]$ so $c \in [a]$ and $c \in [b]$

$\therefore a \sim c$ and $b \sim c$. $\therefore c \sim b$ by symmetry and $a \sim b$ by Transitivity.

Now let x be any element of $[b]$ so $b \sim x$.

$\therefore a \sim b$ and $b \sim x \Rightarrow a \sim x \Rightarrow x \in [a] \quad \therefore [b] \subseteq [a]$.

Similarly let y be any element of $[a]$ so $a \sim y$

$a \sim b \Rightarrow b \sim a$ by symmetry

$b \sim a$ and $a \sim y \Rightarrow b \sim y$ by transitivity

$\therefore y \in [b] \quad \therefore [a] \subseteq [b]$

$\therefore [a] = [b]$

9. (10 pts)

a) (2 pts) The prime factorization for x is $11^5 \cdot 23^4 \cdot 47^2 \cdot 61$ and the prime factorization for the integer y is $7^3 \cdot 11^2 \cdot 13 \cdot 61^5$. What is the prime factorization for the $\gcd(x, y)$?

$$\gcd(x, y) = 11^2 \cdot 61$$

b) (8 pts) Use the Euclidean algorithm to find the $\gcd(589, 323)$ and find integers x and y such that $\gcd(589, 323) = 589x + 323y$.

$$589 = 1 \cdot 323 + 266$$

$$323 = 1 \cdot 266 + 57$$

$$266 = 4 \cdot 57 + 38$$

$$57 = 1 \cdot 38 + 19$$

$$38 = 2 \cdot 19 + 0$$

$$\gcd(589, 323) = 19$$

$$19 = 57 - 38$$

$$= 57 - (266 - 4 \cdot 57) = 5 \cdot 57 - 266$$

$$= 5(323 - 266) - 266 = 5 \cdot 323 - 6 \cdot 266$$

$$= 5 \cdot 323 - 6(589 - 323)$$

$$= 589(-6) + 323(11)$$

$$x = -6 \quad y = 11$$

10. (15 pts) Consider $\mathbb{R}^2 = \{(a,b) : a \text{ and } b \text{ are real numbers}\}$. We can define a relation $\sim$ on $\mathbb{R}^2$ by saying two points (a,b) and (x,y) are related exactly when the difference between the y-coordinates is twice as much as the difference between the x-coordinates. i.e. $(a,b) \sim (x,y)$ if and only if $b-y = 2(a-x)$.

a) (3 pt) List three ordered pairs that are related to $(0,5)$.

$(0,5)$ $(1,7)$, $(2,9)$ Any thing of form $(x, 2x+5)$ $x \in \mathbb{R}$

b) (9 pts) Prove that $\sim$ is an equivalence relation on the set of all ordered pairs $\mathbb{R}^2$.

Reflexive: $\forall (a,b) \in \mathbb{R}^2$ $(a,b) \sim (a,b)$ since $b-b = 0 = 2(a-a)$

Symmetric: If $(a,b) \sim (x,y)$ then $b-y = 2(a-x)$

$$\therefore y-b = -(b-y) = -2(a-x) = 2(x-a)$$

$$\therefore (x,y) \sim (a,b)$$

Transitive: If $(a,b) \sim (x,y)$ and $(x,y) \sim (c,d)$, then

$b-y = 2(a-x)$ and $y-d = 2(x-c)$. Adding these equations we get $(b-y) + (y-d) = 2(a-x) + 2(x-c)$

$$\therefore b-d = 2(a-c)$$

$$\therefore (a,b) \sim (c,d)$$

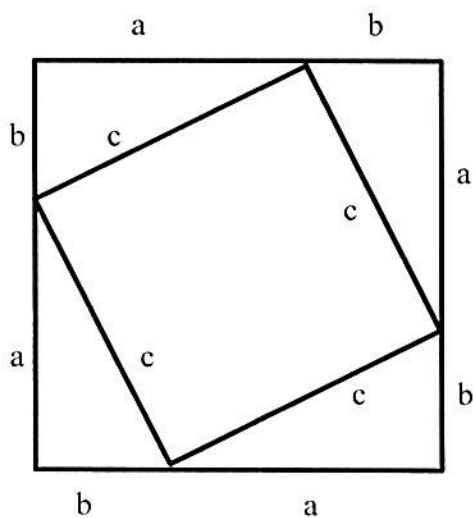
$\sim$ is an equivalence relation since it is reflexive, symmetric and transitive.

c) (3 pts) Give a geometric description of all the points in the equivalence class $[(0,5)]$

$[(0,5)]$ is the set of all points on the line

$$y = 2x + 5$$

11. (6 pts) Explain in complete sentences and equations how this diagram proves the Pythagorean Theorem.



We prove $a^2 + b^2 = c^2$ by computing the area of the large square two ways.

Since side length is $a+b$ we get

$$\text{Area} = (a+b)^2 = a^2 + 2ab + b^2.$$

By decomposing the large square into

4 triangles and the smaller square we get $\text{Area} = 4 \cdot (\frac{1}{2}ab) + c^2 = 2ab + c^2$

$$\therefore a^2 + 2ab + b^2 = 2ab + c^2$$

$$\therefore a^2 + b^2 = c^2$$

12. (9 pts) Let a and b be any integers.

a) Prove that $(a+b)^2 \equiv (a^2 + b^2) \pmod{2}$

Use Test for congruence

$$(a+b)^2 - (a^2 + b^2) = a^2 + 2ab + b^2 - a^2 - b^2 = 2ab \text{ is a multiple of 2.}$$

$$\therefore (a+b)^2 \equiv (a^2 + b^2) \pmod{2}$$

13 b) Prove that $(a+b)^3 \equiv (a^3 + b^3) \pmod{3}$

$$(a+b)^3 - (a^3 + b^3) = a^3 + 3a^2b + 3ab^2 + b^3 - a^3 - b^3 = 3(a^2b + ab^2) \text{ is a multiple of 3}$$

$$\therefore (a+b)^3 \equiv (a^3 + b^3) \pmod{3}$$

c) Prove that it is **not** always true that $(a+b)^4 \equiv (a^4 + b^4) \pmod{4}$

16 If $a=1$ and $b=1$ we get

$$(a+b)^4 = 2^4 = 16 \equiv 0 \pmod{4} \quad \text{But}$$

$$a^4 + b^4 = 1+1 = 2 \equiv 2 \pmod{4}$$